

A
Project Report On

GYM PRO – GYM MANAGEMENT SYSTEM

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By

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CERTIFICATE

(To be provided by the Project Guide on completion of the project)

Preface

This report contains all the outputs of a project undertaken to develop a working model of **GYM PRO – Gym Management System** and to study the existing working system. As part of this study, we analyzed the requirements of the working model and tried to improve it using the technological functionality that the current computer science provides.

This volume contains the documentation summary of the working system, the background study and analysis, the technical requirements and the working specification of the developed system.

We recommend the user to take the advantage of the study presented here and implement the system with their required improvisation. We hope this study will reduce the burden of the reanalysis of the development phase.

Acknowledgment

“It is not possible to prepare a project report without the assistance & encouragement of other people. This one is certainly no exception.”

On the very outset of this report, we would like to extend our sincere & heartfelt obligation towards all the personages who have helped us in this endeavor. Without their active guidance, help, cooperation & encouragement, we would not have made headway in the project. We are ineffably indebted to our HOD / Prof. Jitendra Kabariya for conscientious guidance and encouragement to accomplish this assignment.

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We extend our gratitude to the Department Of Computer Science, Late P.S. Hirapara Mahila BCA College-Jetpur for giving us this opportunity.

Thanking You

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INDEX

• Front Page			
• Certificate			
• Preface			
• Acknowledgement			
Sr. No.	Title		Page No.
1	Project Profile		6
	1.1	Definition	6
	1.2	Scope	6
	1.3	Objective	6
	1.4	Technical Description	6
2	System Analysis and Specification		8
	2.1	Study of Current System	8
	2.2	Weakness of Current System	8
	2.3	Introduction of New System	8
	2.4	Hardware and Software Requirement	8
3	System Requirement Specification		10
	3.1	Proposed System and Advantages	10
4	System Design		11
	4.1	Use – case Diagram	11
	4.2	Data Flow Diagram	15
	4.3	E-R Diagram	15
	4.4	Data Dictionary	16
	4.5	Flowchart (Optional)	21
5	Screen Design		23
6	Testing (Manual, Test Cases and Test Data)		24
	6.1	Manual Testing	24
	6.2	Test Cases	24
7	Enhancements		26
	7.1	Advantages	26
	7.2	Limitation	26
	7.3	Future Scope	26
8	Bibliography		28

1. Project Profile

1.1 Definition

GYM PRO – Gym Management System is a web-based software application developed to manage and maintain the daily activities of a gym. It helps the gym administrator manage member information, membership details, batches, trainers, attendance, payments and other gym-related records in a systematic and efficient way.

The system reduces manual paperwork, saves time, improves data management, and makes it easier to store, update, and retrieve gym information whenever required. It provides three separate, password-protected panels for the Admin, the Trainers and the Members, in addition to a public informational website for visitors.

1.2 Scope

The **GYM PRO – Gym Management System** is designed to manage gym members, memberships, batches, trainers, attendance and payments in an organized way. It reduces manual paperwork and makes daily gym management easier for the owner and staff.

The system helps the administrator to add, update, delete and view records easily. It saves time, reduces errors, and provides quick access to important gym information such as who is due for renewal, which batch is running short of trainers, and how much revenue has been collected.

1.3 Objective

The main objective of the **GYM PRO – Gym Management System** is to develop a simple and efficient system for managing gym activities and records. It aims to reduce manual work and make gym management faster, easier, and more organized.

The major objectives are to:

- Maintain accurate **member and membership records**.
- Manage **batches, trainers, attendance and payment information**.
- Allow a member to join more than one batch while **automatically preventing any clash** between the timings of the selected batches.
- Ensure **no membership becomes Active without a logged or verified payment**, by gating every new registration and renewal behind a payment step.
- Provide easy **adding, updating, approving, deleting and viewing** of records.
- Reduce **paperwork and manual errors**.
- Save time and improve the overall **efficiency of gym management**.

1.4 Technical Description

Technical Aspect	Description
Front End	HTML5, CSS3, Bootstrap 5, Bootstrap Icons, JavaScript, Chart.js, qrcodejs (for the UPI QR code)
Back End	PHP (procedural style, mysqli extension,

Technical Aspect	Description
	prepared statements)
Database	MySQL

2. System Analysis and Specification

2.1 Study of Current System

Most small and medium gyms still manage their day-to-day operations manually. During the study of the existing system, the following observations were made:

- New member details, batch timings and fee amounts are written down in a membership register or note-book.
- Attendance of members is taken on a paper sign-in sheet kept at the reception desk.
- Trainers keep track of which members belong to which batch from memory or hand-written batch sheets pinned on a notice board.
- Renewal of a membership is tracked by checking a physical register or a diary maintained by the gym owner.
- Fee receipts are hand-written and a carbon copy or a simple cash-book is kept for accounts.

2.2 Weakness of Current System

The manual system currently followed by most gyms suffers from the following drawbacks:

- Registers can be lost, damaged, misplaced or duplicated, and there is no backup of the data.
- Checking whether a new admission's chosen time slot clashes with an existing batch is done manually and is error-prone; double-booking of a batch beyond its capacity is common.
- Retrieving a particular member's payment history or attendance record takes considerable time when done manually.
- There is no way for a member to check their own membership status, batch timing, attendance or dues without visiting or calling the gym.
- Sending renewal reminders or announcements to all members has to be done individually by phone or word of mouth, which is slow and unreliable.
- Trainers have no organized way to record attendance or maintain a workout plan for their batch.

2.3 Introduction of New System

GYM PRO – Gym Management System replaces the manual register with a centralized, web-based application. The new system introduces the following improvements:

- A public website lets any visitor view the gym's batches, trainers and gallery, and submit an online registration request; the Admin reviews and approves it, after which the member is created automatically.
- A member can join more than one batch at the same time, and the system automatically prevents the member from picking two batches whose timings overlap.
- Every new registration or renewal is payment-gated: the member is created with a 'Pending Payment' status and cannot reach their dashboard until a payment covering the

amount due is logged by the Admin or verified online — this applies even when the Admin adds or renews a member directly.

- A member can pay online by scanning the gym's own UPI/QR code and submitting the transaction ID, which queues the claim for the Admin to verify before it is treated as a real payment; alternatively, the Admin logs the payment directly when it is collected in person.
- Registration now enforces a minimum age of 18 years, checked both on the date-picker and again on the server.
- Trainers mark attendance for their own assigned batches from their own login, and can maintain a simple workout plan for the batch.
- A dashboard gives the Admin an at-a-glance view of active members, pending requests and revenue collected.

2.4 Hardware and Software Requirement (Feasibility Study)

Software Requirement

Software	Specification
Front End	HTML5, CSS3, Bootstrap 5, JavaScript, Chart.js, qrcodejs
Back End	PHP (compatible with PHP 5.4 as well as PHP 7/8)
Database	MySQL
Web Server	Apache (via XAMPP / WAMP / UwAmp)
Operating System	Windows 10 / 11 (or any OS that can run XAMPP/WAMP)
Browser	Google Chrome / Mozilla Firefox / Microsoft Edge (any modern browser)

Hardware Requirement

Hardware	Specification
Processor	Intel Pentium / Core i3 or above
RAM	4 GB minimum (8 GB recommended)
Hard Disk	100 MB free space for the project and the XAMPP/WAMP installation
Monitor	Standard colour monitor, 1024 x 768 resolution or higher
Input Devices	Keyboard, Mouse

3. System Requirement Specification

3.1 Proposed System and Advantages

Based on the study of the current manual system and its weaknesses, the following project model (modules) is proposed for GYM PRO – Gym Management System:

1) Public Website

Provides Home, About, Batches, Gallery and Contact pages so that any visitor can learn about the gym before joining.

2) Visitor Registration

A visitor can register online by filling in personal details and choosing one or more batches. The system groups batches by time-of-day and blocks any selection whose timings overlap, before the request is sent for approval.

3) Admin Module

The Admin can manage members, trainers and batches; approve or reject registration and renewal requests; log in-person payments and verify or reject online payment submissions; post announcements; and view a dashboard of key statistics.

4) Trainer Module

A trainer can view the batches and members assigned to them, mark daily attendance, maintain a simple workout plan for the batch, and post announcements.

5) Member Module

A member can view their profile, batch(es), attendance and payment history; pay online via UPI/QR and submit the transaction ID when their status is Pending Payment; request renewal of an expiring membership; and view announcements posted by the Admin or trainer.

Advantages of the Proposed System

- Replaces scattered paper registers with a single, centralized, always-available database.
- Automatically prevents a member from being enrolled in two batches whose timings clash, which is not practically possible to check by hand.
- No membership can ever become Active without a logged or Admin-verified payment — the payment gate applies identically whether the member registered themselves or the Admin added them directly, closing a gap that manual systems cannot enforce.
- Lets a member pay online at their own convenience via UPI/QR, while still giving the Admin final control through a Verify/Reject step before any online claim is treated as real money received.
- Cuts down the time needed to search for a member's record, payment history or attendance from minutes to seconds.
- Gives every member and trainer self-service, read-only access to their own information, reducing the number of enquiries handled at the reception desk.
- Scales easily as the number of members, batches or trainers grows, unlike a paper register.

4. System Design

4.1 Use – Case Diagram

The system has four types of actors — Visitor, Member, Trainer and Admin. The use-case diagrams below show the functions available to each actor.

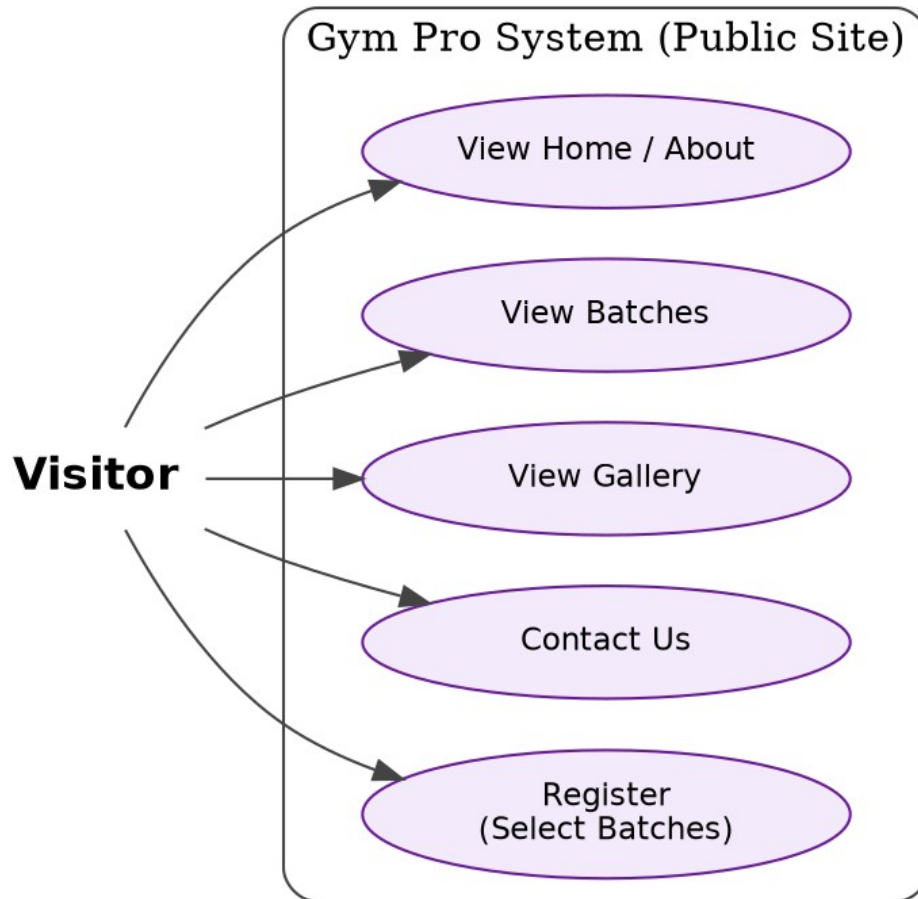


Fig 4.1(a): Use-case diagram – Visitor

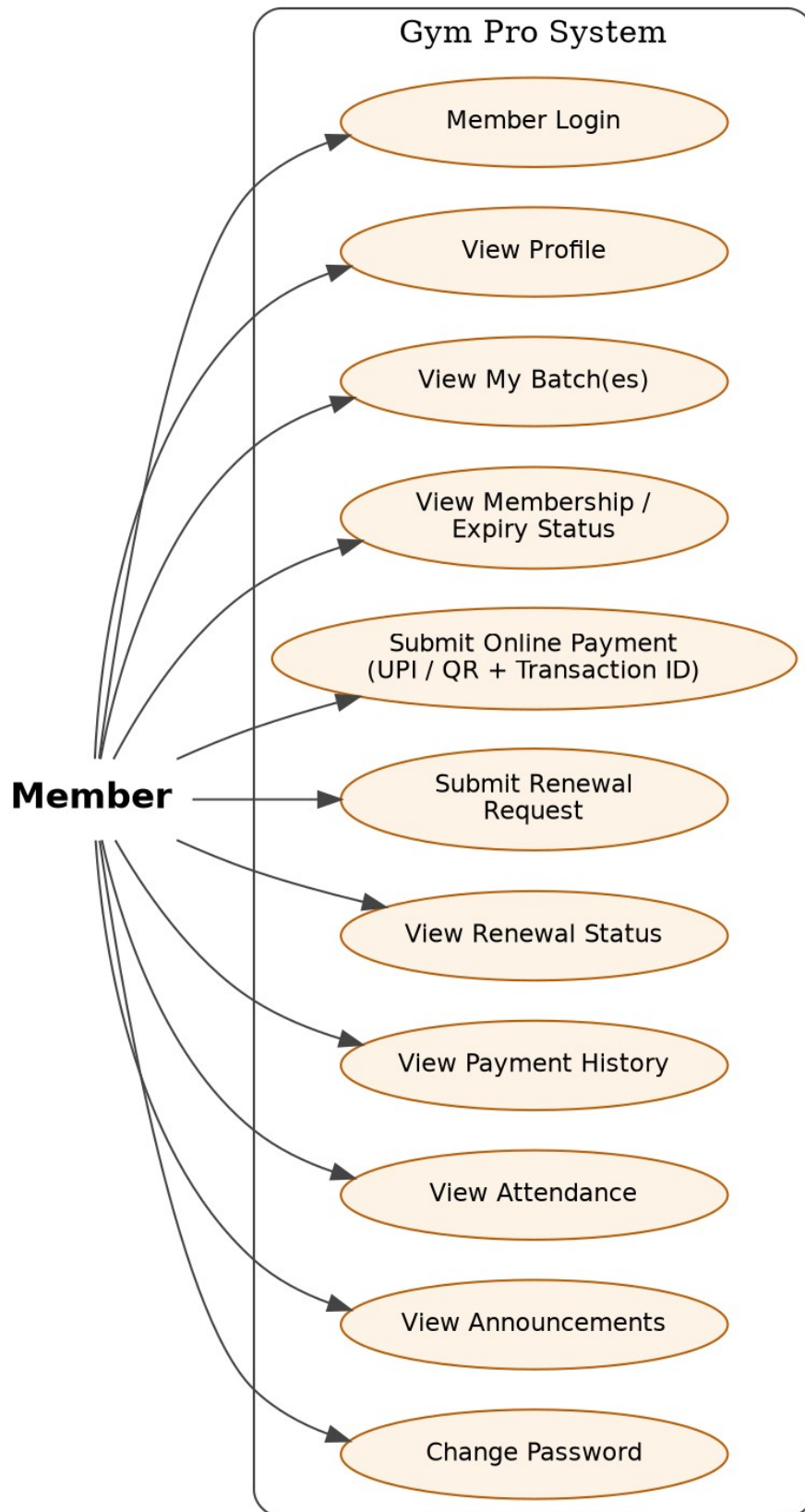


Fig 4.1(b): Use-case diagram – Member

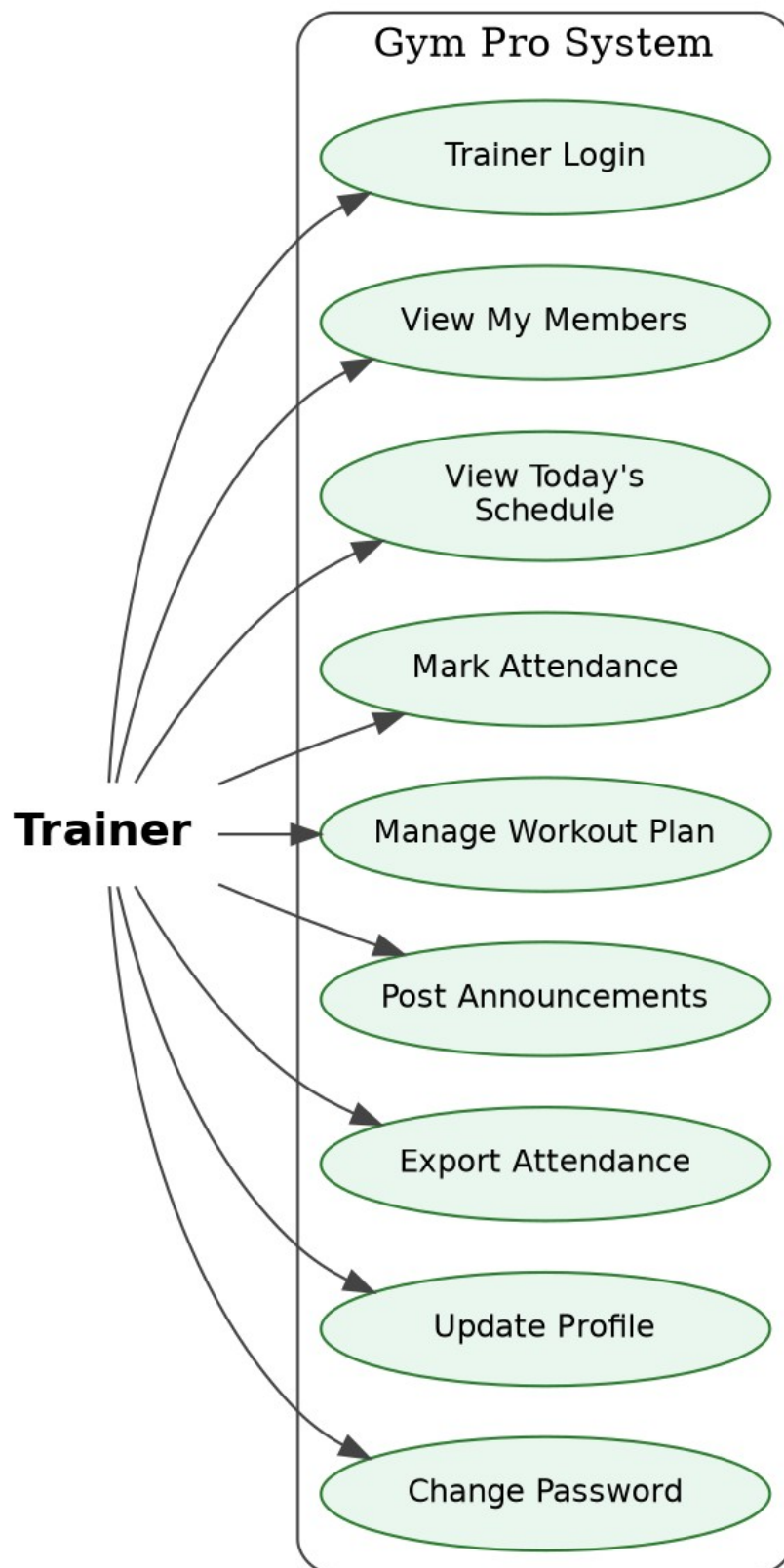


Fig 4.1(c): Use-case diagram – Trainer

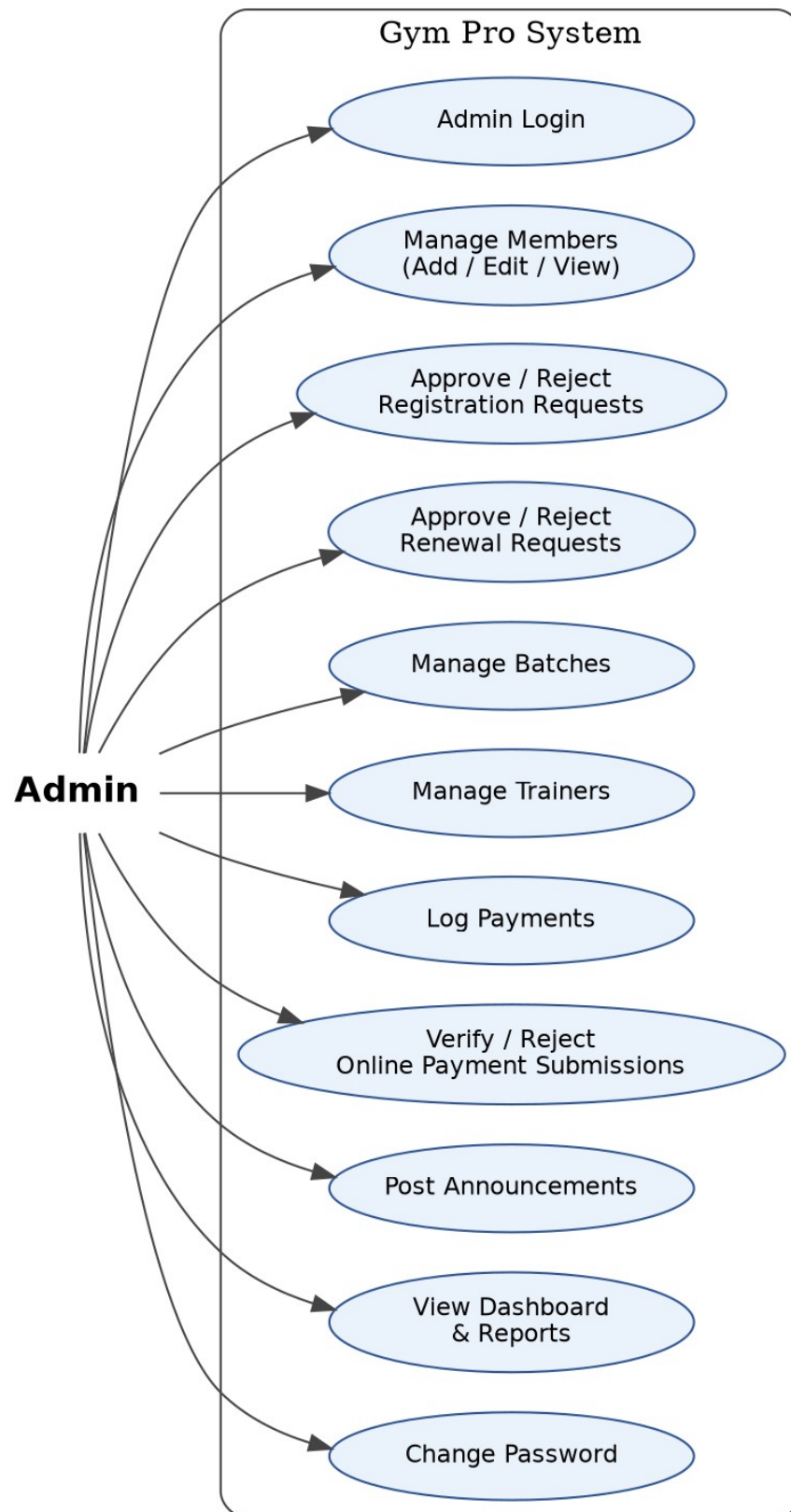


Fig 4.1(d): Use-case diagram – Admin

4.2 Data Flow Diagram

4.3 E-R Diagram

4.4 Data Dictionary

The data dictionary below lists every table of the gym_pro database exactly as created in the system (16 tables in total), showing each column, its data type and its key/constraint.

admins

Column Name	Data Type	Key / Constraint
admin_id	INT (Auto Increment)	Primary Key
admin_username	VARCHAR(50)	Not Null, Unique
admin_password	VARCHAR(255)	Not Null (hashed)

trainers

Column Name	Data Type	Key / Constraint
trainer_id	INT (Auto Increment)	Primary Key
trainer_code	VARCHAR(20)	Not Null, Unique
full_name	VARCHAR(100)	Not Null
gender	ENUM('Male','Female','Other')	Default 'Male'
phone	VARCHAR(20)	Not Null, Unique
email	VARCHAR(100)	Not Null, Unique
specialization	VARCHAR(100)	-
photo	VARCHAR(255)	Default 'default-trainer.png'
username / password	VARCHAR(50) / VARCHAR(255)	Not Null, Unique / Not Null
status	ENUM('Active','Inactive')	Default 'Active'

batches

Column Name	Data Type	Key / Constraint
batch_id	INT (Auto Increment)	Primary Key
batch_name	VARCHAR(100)	Not Null
start_time / end_time	TIME	Not Null
capacity	INT	Default 20
monthly_fee	DECIMAL(10,2)	Default 0
status	ENUM('Active','Inactive')	Default 'Active'

batch_trainers

Column Name	Data Type	Key / Constraint
batch_trainer_id	INT (Auto Increment)	Primary Key
batch_id	INT	Foreign Key -> batches
trainer_id	INT	Foreign Key -> trainers

Column Name	Data Type	Key / Constraint
(batch_id, trainer_id)	-	Unique Key

members

Column Name	Data Type	Key / Constraint
member_id	INT (Auto Increment)	Primary Key
member_code	VARCHAR(20)	Not Null, Unique
photo	VARCHAR(255)	Default 'default-member.png'
full_name	VARCHAR(100)	Not Null
gender	ENUM('Male','Female','Other')	Default 'Male'
dob	DATE	Not Null (18+ enforced)
phone / email	VARCHAR(20) / VARCHAR(100)	Not Null, Unique (both)
address	VARCHAR(255)	-
emergency_contact	VARCHAR(20)	-
duration_months	INT	Default 1
membership_start_date	DATE	Not Null
membership_expiry_date	DATE	Not Null
username / password	VARCHAR(50) / VARCHAR(255)	Not Null, Unique / Not Null
status	ENUM('Active','Expired','Inactive','Pending Payment')	Default 'Active'
created_by	VARCHAR(50)	-
created_method	ENUM('Request','Admin')	Default 'Admin'

member_batches

Column Name	Data Type	Key / Constraint
member_batch_id	INT (Auto Increment)	Primary Key
member_id	INT	Foreign Key -> members
batch_id	INT	Foreign Key -> batches
(member_id, batch_id)	-	Unique Key

payments

Column Name	Data Type	Key / Constraint
payment_id	INT (Auto Increment)	Primary Key
receipt_number	VARCHAR(20)	Not Null, Unique
member_id	INT	Foreign Key -> members
amount	DECIMAL(10,2)	Not Null

Column Name	Data Type	Key / Constraint
payment_date	DATE	Not Null
payment_method	ENUM('Cash','UPI','Card','Bank Transfer','Other')	Default 'Cash'
notes	VARCHAR(255)	-
recorded_by	INT	Foreign Key -> admins

payment_submissions

Column Name	Data Type	Key / Constraint
submission_id	INT (Auto Increment)	Primary Key
member_id	INT	Foreign Key -> members
amount	DECIMAL(10,2)	Not Null (system-calculated)
transaction_id	VARCHAR(100)	Not Null
payment_date	DATE	Not Null
payment_method	VARCHAR(20)	Default 'UPI'
status	ENUM('Pending Verification','Verified','Rejected')	Default 'Pending Verification'
reject_reason	VARCHAR(255)	-
submitted_at	DATETIME	Default Current Timestamp
verified_by	INT	Foreign Key -> admins (nullable)
verified_at	DATETIME	-

requested_members

Column Name	Data Type	Key / Constraint
request_id	INT (Auto Increment)	Primary Key
full_name, gender, dob	VARCHAR/ENUM/DATE	full_name Not Null
phone, email	VARCHAR(20) / VARCHAR(100)	Not Null
address, emergency_contact	VARCHAR	-
duration	INT	Default 1
username / password	VARCHAR(50) / VARCHAR(255)	Not Null
status	ENUM('Pending','Approved','Rejected')	Default 'Pending'
request_date / approved_date	DATETIME	Default Current / Null
approved_by	INT	Foreign Key -> admins (nullable)

requested_member_batches

Column Name	Data Type	Key / Constraint
req_mem_btc_id	INT (Auto Increment)	Primary Key
request_id	INT	Foreign Key -> requested_members
batch_id	INT	Foreign Key -> batches

renewal_requests

Column Name	Data Type	Key / Constraint
renewal_id	INT (Auto Increment)	Primary Key
member_id	INT	Foreign Key -> members
requested_duration	INT	Default 1
status	ENUM('Pending','Approved','Rejected')	Default 'Pending'
request_date / approved_date	DATETIME	Default Current / Null
approved_by	INT	Foreign Key -> admins (nullable)

renewal_request_batches

Column Name	Data Type	Key / Constraint
renwl_req_btc_id	INT (Auto Increment)	Primary Key
renewal_id	INT	Foreign Key -> renewal_requests
batch_id	INT	Foreign Key -> batches

announcements

Column Name	Data Type	Key / Constraint
announcement_id	INT (Auto Increment)	Primary Key
title	VARCHAR(150)	Not Null
description	TEXT	Not Null
publish_date	DATETIME	Default Current Timestamp
posted_by_role	ENUM('Admin','Trainer')	Default 'Admin'
posted_by_id	INT	Not Null

workout_plans

Column Name	Data Type	Key / Constraint
workout_id	INT (Auto Increment)	Primary Key
batch_id	INT	Foreign Key -> batches

Column Name	Data Type	Key / Constraint
day_name	ENUM('Monday'...'Sunday')	Not Null
focus_area	VARCHAR(100)	-
details	TEXT	-
(batch_id, day_name)	-	Unique Key

attendance

Column Name	Data Type	Key / Constraint
attendance_id	INT (Auto Increment)	Primary Key
member_id	INT	Foreign Key -> members
batch_id	INT	Foreign Key -> batches
date	DATE	Not Null
status	ENUM('Present','Absent')	Default 'Present'
(member_id, batch_id, date)	-	Unique Key

settings

Column Name	Data Type	Key / Constraint
setting_id	INT (Auto Increment)	Primary Key
setting_key	VARCHAR(50)	Not Null, Unique
setting_value	VARCHAR(255)	Not Null (e.g. gym_upi_id)

4.5 Flowchart (Optional)

The flowchart below traces registration, approval and the payment gate end to end — a new member always starts as Pending Payment and only becomes Active once a payment is logged by the Admin or verified from an online submission.

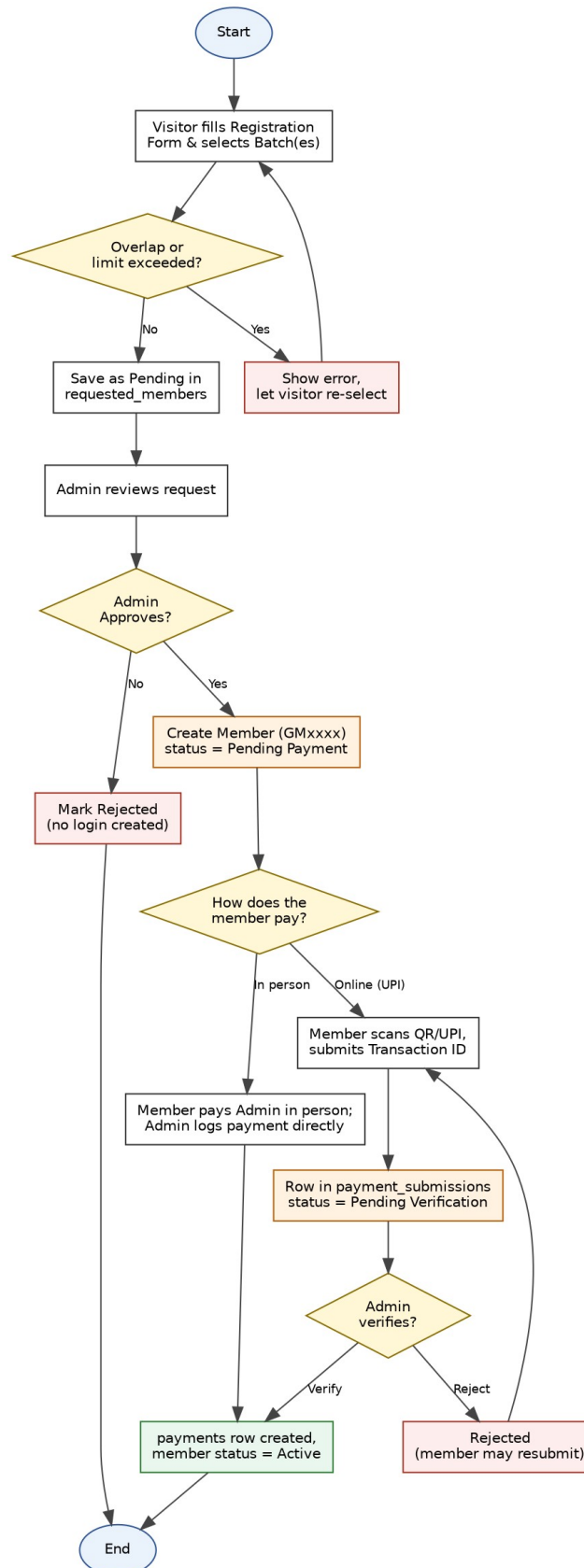


Fig 4.5: Flowchart of Registration, Approval & Payment Gate

5. Screen Design

6. Testing (Manual, Test Cases and Test Data)

6.1 Manual Testing

Every input form of the system was tested manually using the Black Box Testing technique, in which each field is tested without looking at the underlying code — only the input given and the result produced are checked. Four types of input were used for every field:

- Input Type 1 — Valid Input: correct, properly formatted data in every field.
- Input Type 2 — Mandatory Field left blank: a required field is submitted empty.
- Input Type 3 — Invalid Input: data of the wrong format or an out-of-range value.
- Input Type 4 — Optional Field / Default Value / Boundary Condition: an optional field is left at its default, or a value is tested exactly at its allowed limit.

For every module, the mandatory fields are marked "Yes" under Input Type 1, and the overall Pass/Fail result of submitting the form under each of the four input types is recorded in the Result row.

6.2 Test Cases

1. Registration

Field Name	Input Type 1	Input Type 2	Input Type 3	Input Type 4
Full Name	Yes	-	-	-
Phone	Yes	-	-	-
Email	Yes	-	-	-
Date of Birth	Yes	-	-	-
Username	Yes	-	-	-
Password	Yes	-	-	-
Batch Selection	Yes	-	-	-
Result	Pass	Fail	Fail	Pass

Criteria: Full Name, Phone, Email, Date of Birth, Username, Password and at least one Batch are mandatory. Date of Birth must show the visitor is 18 years or older (boundary condition), and the chosen batch(es) must not overlap in timing.

2. Login (Admin / Trainer / Member)

Field Name	Input Type 1	Input Type 2	Input Type 3	Input Type 4
Username	Yes	-	-	-
Password	Yes	-	-	-
Result	Pass	Fail	Fail	Pass

Criteria: Username and Password are mandatory.

3. Renewal Request

Field Name	Input Type 1	Input Type 2	Input Type 3	Input Type 4
Requested Duration	Yes	-	-	-
Batch Selection	Yes	-	-	-
Result	Pass	Fail	Fail	Pass

Criteria: Requested Duration and Batch Selection are mandatory; Duration has a boundary condition of 1 to 12 months.

4. Online Payment Submission

Field Name	Input Type 1	Input Type 2	Input Type 3	Input Type 4
Transaction ID	Yes	-	-	-
Payment Date	Yes	-	-	-
Result	Pass	Fail	Fail	Pass

Criteria: Transaction ID and Payment Date are mandatory. Amount is system-calculated from the member's due fee and cannot be edited (default value).

7. Enhancements

7.1 Advantages

- All database queries use prepared statements (mysqli), which protects the system against SQL-injection attacks.
- Passwords are stored using PHP's password_hash()/password_verify() functions and are never kept in plain text.
- Each panel — Admin, Trainer and Member — has its own independent session-based login; an unauthorized visitor is redirected back to that panel's login page.
- The core business rule — that a member cannot join two batches whose timings overlap — is enforced on the server side, so it cannot be bypassed even if JavaScript is disabled in the browser.
- The payment gate applies with no exceptions: every path that creates or renews a member, including the Admin's own Add Member and Renew Member actions, lands in Pending Payment first, so no membership can become Active without a logged or verified payment.
- An online payment is never trusted on its own — the member only submits a transaction ID, and nothing is treated as a real payment until the Admin verifies it, which keeps a human check in the loop.
- A trainer can view and manage only the batches actually assigned to them, not the entire gym's data.
- The system is built to run on both modern PHP (7.x / 8.x) and older PHP 5.4 installations, which is useful in college lab environments.

7.2 Limitation

- There is still no real payment gateway integration; online payment is a submit-and-verify claim against the gym's own UPI ID, not an automated checkout.
- Photo upload for members and trainers is not yet implemented, although the database already has a column reserved for it.
- No automatic email or SMS notification is sent when a registration or renewal request is approved or rejected, or when a payment is verified — the member has to log in to check the status.
- Long lists (members, trainers, requests) do not yet have pagination, which may slow down as the number of records grows very large.
- The Contact Us form on the public site only shows a confirmation message; enquiries are not yet stored in the database for the Admin to review later.

7.3 Future Scope

- Add real photo upload for members and trainers using PHP's move_uploaded_file() function.
- Integrate an actual payment gateway (for example Razorpay) so an online payment is confirmed automatically instead of waiting for Admin verification.

- Send an automatic email or SMS notification when a registration/renewal request is approved or rejected, or when a payment submission is verified or rejected.
- Add a contact_messages table and an Admin inbox so enquiries from the Contact Us page can be tracked and answered.
- Add pagination and search/filter options to all the long list pages.
- Package the member-facing pages as a Progressive Web App (PWA) so members can check their schedule, attendance and payment status conveniently on mobile.

8. Bibliography

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<https://www.w3schools.com>

<https://getbootstrap.com>

<https://dev.mysql.com/doc/>

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<https://davidshimjs.github.io/qrcodejs/>

Books:

- PHP and MySQL Web Development – Luke Welling & Laura Thomson
- Learning PHP, MySQL & JavaScript – Robin Nixon